COMMENTARY

Uncrewed Platforms Have Been Critical to Ukraine's Success in the Black Sea

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10 Minute Read

A consideration of other options shows that the success Ukraine has had in keeping the Russian Black Sea Fleet at bay could only have been achieved using uncrewed vessels.

The Russo-Ukrainian War may mark the beginning of a new era in naval drone warfare. In **October 2022**, small surface drones breached Russian defences and drove deep inside Sevastopol harbour, even targeting the Russian Navy flagship, foreshadowing a new phase in the war. Since that initial high-profile mission, Ukraine's naval drones have played pivotal role in reshaping the conflict in the Black Sea.

The success of Ukraine's uncrewed surface vessels (USVs) is indisputable. The Russian Navy is now on the defensive, and consequently conducting minimal naval operations. It is one of the clearest examples of the success of asymmetrical threats against a larger, more powerful, opponent. The popular meme 'Russia has lost another warship to a country without a navy, in a land war' contains some truth. The Ministry of Defence **recently** put it more soberly, noting that 'The size of the Russian Black Sea Naval Fleet has proven no match for Ukraine's maritime innovations'.

This remarkable shift, where the hunter becomes the prey on a grand scale, offers numerous lessons for navies both large and small. Crucially, this achievement has only been possible due to advances in uncrewed technology. Similar results would not have been possible if Ukraine had relied on crewed platforms.

The Drone Navy's Success

Much has been written about Ukraine's remarkable naval success in the Black Sea. While the caveats that this is an ongoing war and that the situation could change at any moment apply, it is indeed incredible. Russia has lost sea control, and its fleet is for the most part holed up behind layers of boom defences in ports as far from Odesa as possible.

At the start of the February 2022 full-scale invasion the Black Sea could, without exaggeration, be described as a Russian (and Turkish) lake. Other naval powers, not least the Ukrainian Navy, had minimal control. Certainly, short of direct Turkish intervention, Russia enjoyed almost complete freedom of operation.

Ukraine's flagship, the *Krivak-III*-class frigate *Hetman Sahaidachny*, was **scuttled at its berth** in Mykolaiv as the Russian advance drew close. Ships which did take on the Russians directly did not fare well. The patrol ship *Sloviansk*, which lacked significant air defences, was **sunk on 3 March 2022** by Russian aircraft. Ships in Ukraine's ports in the Sea of Azov were destroyed or captured as Russian forces advanced.

Thus, within days of the invasion the Ukrainian fleet was all but non-existent. Meanwhile, the Russians enjoyed complete dominance, able to sail off the Odesa coast with apparent impunity.

> **There are clear signs that Ukraine is exploring automation, including for the 'last mile' of an attack where electronic warfare may interfere with communications**

This was to change in April 2022 when two of Ukraine's Neptune anti-ship missiles ripped through the port side of the Black Sea Fleet's flagship, the *Slava*-class cruiser *Moskva*. The ship **subsequently sunk.** This represented the first significant change, with Ukrainian anti-ship missiles (both the homegrown Neptune and Western-supplied Harpoon) driving the Russians back from the western side of the northwest Black Sea. But the land-based missiles could only threaten the Russian Navy so far. Threatening them further south and east, and ultimately throughout the entire sea, required boats.

These boats were USVs – small craft with a camera, satellite communications, and a warhead in the bow. Specifications vary, but they are generally the type of craft which can be built in a garage or small industrial unit.

The main technological enabler has been high-capacity two-way satellite communications such as Starlink and Kymeta, both of which have been observed in use. These allow human-in-the-loop navigation and targeting throughout the entire mission. Humans have proven adaptable to the tactical situation and faster to deploy than fully automated equivalents. There are clear signs that Ukraine is exploring automation, including for the 'last mile' of an attack where electronic warfare may interfere with communications. But communications were key precisely because Ukraine did not have to rely on automation from day one.

USV operations ramped up between the spring and summer of 2023, reaching a climax when the *Ropucha*-class landing ship *Olenegorsky Gornyak* was **struck by one outside Novorossiysk.** The ship was seriously damaged, only just avoiding sinking by being towed back to port. The *Olenegorsky Gornyak* survived, but others were less fortunate. Since then, another landing ship, a missile corvette, a patrol ship, a minesweeper and several landing craft and smaller boats have all been sunk or destroyed by USVs.

It is not just ships being sunk that has had a tangible impact. The unsuccessful attacks, and mere threat of attack, have contributed. The Russian Navy has moved almost all its major assets to the port of Novorossiysk on the eastern coast of the Black Sea, rerouted its 'Syrian Express' supply lines to no longer use the Black Sea, and generally minimised patrolling.

Ukraine has been building large numbers of these USVs – over 100 according to public information. The ones involved in offensive missions – at least the ones publicised – are operated by the GUR (Defence Intelligence of Ukraine) and SBU (Security Service of Ukraine), as opposed to the traditional navy. The GUR are known for the Magura drones, while the SBU are associated with the Sea Baby type. Both organisations have enjoyed significant successes and contributed to the overall situation.

Could This Have Been Achieved with Crewed Platforms?

It is difficult to conceive of a way in which the same effect could be achieved with crewed platforms – in particular, the ability to put the Russian fleet at risk throughout the entire Black Sea, including close to the Russian coast where the full might of the Russian military can be brought to bear.

The Conventional Navy Option

If we look at the February 2022 Ukrainian Navy, with one frigate and various gun-armed boats, there was nothing that could survive in open water against the Russians. Even if we imagine the arrival of the **two MILGEM corvettes** under construction in Turkey, the picture doesn't change.



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These vessels are thoroughly modern and have respectable point air defence and anti-submarine capabilities, but they are too few and, on their own, too vulnerable. While individually high-quality, they are exactly the type of navy that the Russians were geared up to fight. The moment they arrived in Ukraine you could expect them to be high-priority targets for the Russian Navy. They'd face a formidable wall of high-performance anti-ship missiles launched from ships, shore, aircraft and submarines. In port they'd be vulnerable to cruise missiles and stand-off bombs, and as they set sail, they could quickly face torpedoes. The one thing Russia isn't short of is anti-ship missiles.

This highlights the first major advantage of the USVs. Their small size means that they are not particularly vulnerable to anti-ship cruise missiles, while their speed and size provide decent defence against torpedoes. They are less vulnerable in port because they need minimal shore infrastructure to launch. There is no need to tie them up in port in a predictable way. Cruise missiles and stand-off weapons can potentially target their storage sites, but this is less of a threat than when they are tied up in a naval base.

Their one major vulnerability is from the air. Efforts have been made to address this, with air-defence missiles added, but the challenge remains. Yet this has to be kept in perspective. Their vulnerability decreases their effectiveness but hasn't stopped them from performing thus far. And whatever their vulnerability, it is still less than that of most surface warships.

The Mosquito Fleet Option

Ukraine has been able to build up its small craft, with numerous types supplied by Western governments. These now have the capability to patrol the coast and conduct amphibious raids and special operations. At the moment their impact is restricted to the northwestern Black Sea.

Amphibious raiding has proven valuable in northwestern Crimea, but has limited reach due to human endurance. Special forces missions – possibly involving underwater vehicles – have some potential, but these are isolated attacks. And none of this threatens a Russian ship in open waters.

Could the remaining mosquito fleet of assault boats, patrol craft and inflatables possibly be used in a similar way to the USVs? The problem is that the craft lack the endurance, and are too vulnerable, for sustained operations in the eastern Black Sea. And it is not just the vessels themselves that lack the endurance; their human crews would be stretched. The craft would need to carry more crew and have better accommodation and facilities, which would quickly result in bigger craft.

Uncrewed vessels can be inexpensive so they do not need to return to base, effectively doubling their operational radius. Crewed vessels, meanwhile, would need some form of stand-off attack capability to allow them to return home after an attack, unlike the USVs which sacrifice themselves for the kill. The combination of needing half the fuel and no crew accommodation also means

that the USV can be much smaller, so much so that detection on a ship's radars becomes a challenge.

> **Anti-submarine warfare is inherently difficult**

Crewed vessels would share the USVs' vulnerability to Russian aviation. They could possibly be better defended by short-range missiles, but this advantage is likely marginal. And being larger, they would be much more vulnerable to traditional anti-ship missiles. The missions would thus be very high-risk for the crews, even suicidal. USVs, meanwhile, are expendable and attrition can be factored in without serious concern for crew morale or effectiveness. Ukrainian USVs are working at a ratio of around 10:1 losses to kills, but that's still in their favour.

Creating A Submarine Threat

Perhaps the most promising alternative would be crewed submarines – possibly if recently retired or soon-to-be-retired conventionally powered submarines were transferred from Western navies, or new-build boats were acquired. Submarines could offer a stealthy, deadly and persistent threat throughout the Black Sea.

Although the Russian Black Sea Fleet has, on paper, considerable anti-submarine assets, it is questionable how well they could deal with them. Anti-submarine warfare is inherently difficult. A lot of the Black Sea Fleet's equipment is dated, and readiness levels are open to question. While one wouldn't expect the Black Sea Fleet to sit back and let Ukraine dominate, this plays to the Russians' weaknesses.

However, there are also major limitations which bring this scenario into question. Delivering the submarines would be the first challenge, since the Bosporus has been closed to Ukrainian and Russian combatants and submarines. Next, the submarines would have to spend a lot of time in port. They would, like frigates, be prime targets while doing so. Another challenge would be crewing. This also applies to surface combatants, but it's greater with submarines. Training could be expedited, but it would still take many months. It is questionable whether Ukraine would have been able to train the crews in anything like the same timeframe in which the USV operations have taken place. There are still shore crew to be trained, including in the new set of drone piloting, but the time required is competitively less.

Conclusion

Ukraine's success in pushing Russia's Black Sea Fleet back to Novorossiysk is in no small part down to the use of USVs. Together with the shore-based anti-ship missiles – later joined by **Storm Shadow and ATACMS missiles** – which threaten ships in Crimean ports, USVs have denied Russia sea control. And their viability stems directly from the fact that they are uncrewed.

This piece has not explored the limitations of these craft, or how they might evolve in the future – particularly as Russia becomes more adept at countering them. But that would not change the story. It is difficult to imagine any alternative means by which Ukraine could have achieved the same aims.

It is equally difficult to imagine these vessels being anything like as effective if the full-scale invasion had occurred just a few years earlier, before the latest satellite communications like Starlink and Kymeta were available. In this one small aspect, the timing was perfect for Ukraine

Navies and planners will no doubt be looking at the war closely. It redraws the threat picture for larger navies that are looking to prepare for future

operations. And for countries facing similar threats, uncrewed platforms offer significant advantages. The age of naval drone warfare has arrived.

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